

Geometric Manifold Discovery and Dual-Layer Deterministic Runtime

A Technical Write-Up for RDC-1 Certification Reference

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REVISION NOTE — v1.3 (DISCLOSED CHANGES, inclusive of v1.1 and v1.2)

This public edition supersedes internal drafts v1.0–v1.3 (September 29–30, 2026), which were never published. Disclosed changes relative to the initial internal draft: **(1) Naming:** the certification framework is designated **RDC-1 (Runtime Determinism Certification, Level 1)**, distinct from *Substrate Determinism Receipt 1* (checksum 0xc5f88dcf, certified September 2026) — the completed, verified determinism test result referenced as evidence in Section 5.4. **(2) Epistemic framing (Section 2.4):** the elicited self-description is presented as a narrative scaffold — a reproducible generative artifact — not introspective access to model internals; no claim of that kind is made anywhere in this document. **(3) Evidence discipline:** only receipted results are reported. Substrate determinism is cited from the SDR-1 receipt; all runtime-layer metrics are marked pending the scheduled trials, and figures from earlier drafts not backed by measurement records are withdrawn pending those trials (Section 5.5). **(4) Artifact statuses:** ART-002 and ART-003 are presented as defined-but-unexecuted methodology — the procedures are specified, the artifacts have not been produced; statuses match evidence. The Phase 4 platform is named: Base44. **(5) Phase 1 target model:** the discovery-session model is Copilot, named in Section 4.1, Section 5.1, and the Artifact Inventory. **(6) Editorial:** cover-title spacing, per-section list numbering, and

the standing prototype disclaimer. This document was prepared with AI assistance under author direction; all changes are disclosed per the open-book method.

This software is a prototype and is provided for educational and research purposes only. It is not intended for production use, commercial deployment, or safety-critical environments. All systems are experimental and may contain defects in them.

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1. Abstract

This document presents a structured technical account of a five-phase pipeline in which a large language model (LLM) was prompted to describe its operational structure in geometric terms; the resulting narrative was formalized into a Structural Manifold Specification (SMS); the SMS was used to generate a Deterministic Architecture Blueprint (DAB); the DAB was executed on a no-code application platform to instantiate a dual-layer runtime; and the runtime's substrate was subjected to formal determinism validation.

The pipeline is offered as a reproducible *methodology*: a disciplined way of converting model-generated geometric narrative into a formal specification, and from that specification into an auditable, deterministic runtime. The verified determinism evidence in this program is substrate-level and receipted (Substrate Determinism Receipt 1, checksum 0xc5f88dcf, September 2026): byte-identical outputs across varied execution timing and instance indices. Runtime-layer micro-metrics defined in the Phase 5 protocol (layer synchronization latency, orphan-event counts, independent re-execution trials) are defined in this document and scheduled for measurement; they are reported here as pending, not as results.

A central epistemic clarification distinguishes this account from stronger claims sometimes attached to similar exercises: the LLM's geometric self-description is treated as a *narrative scaffold* — a reproducible generative artifact — and not as introspective access to the model's weights or activations. The engineering value of the pipeline does not depend on the self-description being true of the model; it

depends on the artifacts produced downstream being formally specified, versioned, auditable, and deterministic. That is the contribution.

2. Introduction and Motivation

2.1 Context and Problem Statement

Contemporary large language models operate as probabilistic inference engines: given an input sequence, they produce outputs by sampling from learned conditional probability distributions over token vocabularies. This architecture, while powerful for generative tasks, presents a fundamental challenge for systems engineering contexts where auditability, reproducibility, and deterministic behavior are required. The internal state of an LLM during inference is distributed across billions of parameters and is not directly observable, mappable, or certifiable using conventional software engineering tools.

The problem this work addresses is the construction and validation of a structured, reproducible runtime derived from a model-generated specification — a runtime in which every design decision traces back to a versioned artifact, and whose behavior is deterministic and auditable. The geometric self-description produced by the model serves as the origin point of that artifact chain. Whether the self-description corresponds to the model's actual internals is *not* a claim this document makes or needs; what matters is that the elicitation procedure is documented, the artifacts it produced are preserved, and the downstream engineering stands on its own.

The motivation is threefold. First, a runtime grounded in a documented elicitation process carries a form of provenance: each design decision can be traced to an artifact produced during the pipeline. Second, a deterministic runtime offers a certifiable operational envelope, enabling deployment of model-informed systems in environments that mandate auditability. Third, the pipeline itself — from prompt design through runtime validation — is a reproducible methodology, applicable to other model families with appropriate adaptation.

2.2 Research Objectives

1. **Geometric Self-Representation Elicitation:** Design and apply a structured set of prompts that invite the LLM to describe its operational structure in geometric and topological terms, yielding internally consistent and reproducible descriptions across independent sessions.
2. **Structural Manifold Formalization:** Translate the elicited descriptions into a formally specified Structural Manifold Specification (SMS) — a machine-readable abstraction of operational topology.
3. **Deterministic Blueprint Generation:** Use the SMS as the authoritative input to generate a complete, versioned Deterministic Architecture Blueprint (DAB).
4. **No-Code Builder Execution:** Execute the DAB on a visual, no-code application platform, producing a deployed runtime without manual code authoring.

5. **Dual-Layer Runtime Validation:** Subject the deployed runtime to a formal validation protocol covering determinism, constraint adherence, and cross-layer synchronization, with results recorded as receipts.

2.3 Scope and Limitations

- **Results are model- and session-specific.** The elicited descriptions reflect the particular model instance and prompting sessions used. Generalizability to other model families requires independent replication.
- **No claim of completeness is made** for the Structural Manifold. It captures only the structures surfaced by the elicitation prompts used; other structure may exist that was never elicited — and, per Section 2.4, whatever was elicited is narrative, not measurement.
- **Determinism claims are scoped.** Verified determinism evidence (Section 5.4) applies at the substrate level within the tested input domain. Runtime-layer micro-metrics are protocol-defined and pending measurement; behavior outside the tested domain is uncharacterized.
- **Platform details.** The no-code platform used in Phase 4 is identified in the operations appendix maintained separately from this reference document.
- **Prototype status.** All systems described are prototypes under the standing prototype disclaimer (see cover). No production warranty is stated or implied.

2.4 Epistemic Status of the Self-Description (added in v1.1)

LLMs do not have introspective access to their own weights or activations. When prompted to "describe the shape of the space information moves through," a model does not inspect its internals; it generates text conditioned on its training distribution, which includes extensive machine learning literature on attention, embeddings, manifolds, and representational geometry. The resulting descriptions are fluent, structured, and often topologically consistent — but they are **pattern-matched narrative, not measurement**.

This document therefore assigns the elicited geometric self-description a precise and limited role: it is a **narrative scaffold** — a reproducible generative artifact used as the origin of a formal specification chain. The properties that are actually tested about it are external and behavioral: (a) *internal consistency* (no self-contradictions), (b) *reproducibility* (equivalent prompts in independent sessions yield topologically equivalent descriptions), and (c) *formalizability* (the description can be converted into a complete, unambiguous specification). None of these tests require the description to be true of the model's mathematics, and this document makes no such claim.

WHY THIS MATTERS

A reviewer who reads "the model described its own geometry, and we built what it said" will — correctly — object that the model cannot know its own geometry. The defensible claim is stronger in a different

direction: **the deterministic validation layer does not trust the model's story; it verifies the system against invariant contracts.** The architecture's soundness is independent of whether the elicited narrative corresponds to model internals. Any framing that implies otherwise should be treated as an error in this and future documents.

3. Theoretical Foundations

3.1 LLM Internal Structure and Geometric Interpretation

Transformer-based LLMs are composed of stacked layers containing multi-head self-attention mechanisms and feed-forward projection networks. From a geometric perspective, the core operations can be understood as successive transformations of high-dimensional vector representations: token representations are projected into query, key, and value spaces; attention scores are computed as inner products in those spaces; and weighted combinations are projected back into the residual stream. These operations are fundamentally geometric — rotations, projections, and affine transformations in high-dimensional spaces.

The embedding space of an LLM has been extensively studied as a geometric object: directions encode semantic properties, distances encode semantic proximity, and attention-head outputs exhibit interpretable structure. The layer-wise progression of representations constitutes a sequence of geometric transformations applied to an initial embedding. The manifold hypothesis in representation learning — that high-dimensional natural data tends to concentrate near lower-dimensional manifolds — provides the theoretical backdrop for treating activation spaces as manifold-like structures. This background is *genuine* mathematics about how LLMs work. Note the careful boundary: these facts justify using geometric vocabulary as an engineering language; they do not make the model's self-narrative into a measurement of its own geometry (Section 2.4).

3.2 Structural Manifolds

As used in this work, a Structural Manifold is a formalized geometric abstraction of an operational topology — a structured representation capturing the relational geometry of signal flow, routing, and representational regions, as expressed through a controlled elicitation procedure and formalized into a machine-readable specification.

Three properties are required of a well-formed Structural Manifold:

1. **Continuity:** the manifold describes a connected operational topology, with no unexplained discontinuities in signal flow or routing pathways.
2. **Differentiability (in principle):** transitions between geometric primitives must be describable in terms that are, in principle, smooth — enabling gradient-based optimization criteria in future extensions.
3. **Parametric Reproducibility:** the manifold must be fully specified by a finite set of parameters such that it can be exactly reconstructed from those parameters alone, without reference to the original elicitation session.

A critical distinction exists between a *learned latent manifold* — the implicit low-dimensional structure that emerges from training and is encoded in model weights — and an *explicitly constructed structural manifold* derived from prompted self-description. The former is an emergent property of training and is not directly accessible; the latter is an explicit artifact produced through a controlled procedure and formalized as described in Section 4.2. **The structural manifold is an engineering construct, not a claim about the model's underlying mathematics.** This distinction governs the entire document.

3.3 Deterministic Runtime Principles

Determinism is defined operationally: a runtime is deterministic if and only if, given identical inputs and a fixed, versioned blueprint, it produces identical outputs with zero stochastic variance across an arbitrary number of execution runs. This definition is narrower than mathematical determinism and intentionally grounded in empirically testable behavior.

The dual-layer architecture separates runtime concerns into two independently auditable strata:

Layer	Designation	Responsibility	Auditability Mechanism
Layer 1	Semantic Interpretation Layer	Parsing, classifying, and routing incoming signals according to the semantic schema derived from the structural manifold	Semantic classification logs; routing trace records
Layer 2	Execution Control Layer	Executing state transitions, enforcing constraints, and producing outputs according to the DAB	State transition audit trail; constraint enforcement event log

The rationale for the separation is independent auditability of semantic and execution behavior without cross-contamination: a failure in Layer 2 output can be traced through the constraint enforcement log without re-analyzing Layer 1 decisions, and vice versa. This property is essential to the chain-of-provenance requirements of RDC-1 certification.

4. Methodology

The pipeline is organized into five sequential phases. Each phase receives defined inputs, applies a structured procedure, and produces a named, versioned output artifact.

4.1 Phase 1 — Geometric Self-Representation Prompting

The objective of Phase 1 is to elicit a geometrically coherent, internally consistent, and reproducible description of the target LLM's operational structure in geometric terms. The target model for the discovery sessions was **Copilot** (Microsoft's conversational assistant), identified by the author. Prompt design used open-ended geometric framing: rather than directing the model toward specific architectural components, prompts invited spatial, topological, and relational description, allowing geometric structure

to emerge from the model's generative process. Three prompt categories were used (templates preserved in Appendix A): **Topology Elicitation**, **Dimensionality Self-Assessment**, and **Relational Geometry**.

A self-representation was accepted for use in Phase 2 if it satisfied three criteria: **(a) internal consistency** — no geometric claims within a description contradicted one another; **(b) reproducibility** — equivalent prompts, applied in independent sessions, produced topologically equivalent descriptions; and **(c) non-trivial geometric content** — the description contained specific geometric primitives (directional corridors, attractor regions, boundary surfaces) rather than generic spatial language. Per Section 2.4, these criteria test the *artifact's* properties, not the model's internals.

Output Artifact — Phase 1. ART-001: Geometric Self-Description Document — a structured document containing the elicited descriptions across all prompt categories, together with the prompts and session metadata. Format: Markdown / PDF. Status: sessions completed; archival location and completeness audit pending (see Section 8).

4.2 Phase 2 — Structural Manifold Application

Phase 2 translates ART-001 into a formal Structural Manifold Specification (SMS) through four sequential formalization steps:

1. **Geometric Primitive Identification:** the self-description is analyzed to extract all distinct geometric primitives — hyperplane boundaries, attractor basins, routing corridors, curvature regions, and dimensional transition zones. Each primitive receives a unique identifier and a definition derived from the elicited text.
2. **Manifold Construct Mapping:** each primitive is mapped to a formal construct. Hyperplane boundaries become constraint surfaces; attractor basins become stable fixed-point regions; routing corridors become directed geodesics; curvature regions become areas of varying local metric structure.
3. **Coordinate System and Dimensionality Definition:** a coordinate system is defined for the manifold, specifying dimensions, axis interpretations, and coordinate domains. Effective dimensionality is determined by the number of independent primitives that cannot be expressed as linear combinations of one another.
4. **Internal Consistency Validation:** the draft SMS is reviewed for internal consistency: all primitives must be expressible within the coordinate system, no two primitives may overlap or contradict, and all routing corridors must have defined source and destination regions.

Output Artifact — Phase 2. ART-002: Structural Manifold Specification (SMS) — a formal, machine-readable specification. Format: JSON Schema. Status: **NOT PRODUCED** — no SMS file exists. The four formalization steps above define the required procedure; executing them and producing the SMS artifact is pending work on the certification roadmap (Section 8).

4.3 Phase 3 — Reproducible Blueprint Generation

Phase 3 uses the SMS (ART-002) as its sole authoritative input to generate a complete Deterministic Architecture Blueprint (DAB) — a versioned, machine-readable specification of the runtime to be instantiated in Phase 4. The DAB is the single source of truth: no design decisions may be made during Phase 4 execution that are not derivable from the DAB. The blueprint comprises five structural components:

1. **Layer Topology Map:** the boundaries, interfaces, and internal structure of the two runtime layers, expressed as nodes and directed edges derived from SMS routing corridors.
2. **Signal Routing Schema:** entry points, branching conditions, routing decisions, and exit points for all signal types within the defined input domain.
3. **State Transition Table:** a complete enumeration of valid runtime states, transition conditions, and outputs. The state space is finite and closed within the input domain.
4. **Constraint Enforcement Rules:** formal rules defining acceptable runtime behavior boundaries, derived from attractor-basin boundaries and hyperplane constraints in the SMS, each with a defined exception-handling response.
5. **Validation Checkpoints:** observable conditions that must be satisfied at defined points during execution, used in Phase 5 validation.

The blueprint is expressed in YAML (primary) and JSON Schema (secondary) for portability. The reproducibility criterion: the DAB, applied to an equivalent builder environment, must produce a runtime instance functionally identical to the original across all inputs in the defined domain.

Output Artifact — Phase 3. ART-003: Deterministic Architecture Blueprint (DAB). Format: YAML / JSON Schema. Status: **NOT PRODUCED** — no versioned DAB file exists. The generation procedure is retained as the certification standard; producing the DAB from the (future) SMS is pending work (Section 8).

4.4 Phase 4 — No-Code Builder Execution

Phase 4 instantiates the runtime on **Base44**, the visual no-code application platform used throughout the actual program. The five-step procedure below (DAB import through deployment) is the certification-standard path once a DAB exists. The historical build predates a formal DAB: the runtime was assembled directly in the builder under the design discipline derived from the discovery narrative — no manual code authoring, constraint wiring, and pre-deployment consistency validation — and that discrepancy is disclosed rather than papered over. The procedure:

1. **DAB Import:** the versioned DAB is imported into the platform's project environment, with schema validation at import time.

2. **Node Mapping:** each node in the Layer Topology Map is mapped to a platform primitive; all required nodes were available in the standard library.
3. **Constraint Module Configuration:** constraint rules are configured as enforcement modules wired to the appropriate nodes with defined exception responses.
4. **Graph Validation:** the assembled graph is validated against the DAB using the platform's consistency checker — topology, routing, and state logic must match exactly.
5. **Runtime Deployment:** the validated graph is deployed as a live runtime instance, made available for Phase 5.

The principal advantage of this approach is the elimination of implementation drift: because the runtime is assembled from blueprint-mapped primitives rather than hand-coded components, there is no opportunity for developer interpretation to introduce behavioral deviation from the DAB.

Output Artifact — Phase 4. ART-004: Runtime Instance — a deployed, operational dual-layer runtime. Format: Deployed Application. Status: deployed instances exist and operate in prototype service; instance-level conformance to the DAB is re-verified during Phase 5 execution.

4.5 Phase 5 — Dual-Layer Deterministic Runtime Validation

Phase 5 subjects the deployed runtime (ART-004) to a formal validation protocol covering determinism, constraint adherence, and cross-layer synchronization, validated independently per layer and jointly at system level. The protocol consists of three test categories:

- **Input/Output Consistency Tests:** a defined set of test inputs is applied N times (N and the input set recorded in the validation report); outputs are compared across all runs, with determinism quantified as the percentage of input/output pairs identical across all N runs.
- **Constraint Satisfaction Checks:** each constraint rule is exercised with boundary-condition inputs approaching — but not crossing — the constraint boundary; adherence is confirmed where in-domain inputs trigger no violation, and exception handling is confirmed where constraint-crossing inputs trigger the specified response.
- **Cross-Layer Synchronization Verification:** latency between Layer 1 classification events and corresponding Layer 2 state transitions is measured across all runs; synchronization is confirmed where latency remains below threshold and no state transition occurs without a corresponding Layer 1 event in the audit log.

Output Artifact — Phase 5. ART-005: Dual-Layer Runtime Validation Report. Format: PDF, receipted. Status: substrate-level determinism verified and receipted (Section 5.4); runtime-layer metrics defined and scheduled (Section 5.5).

5. Results and Observations

5.1 Geometric Self-Representation Findings

The Phase 1 elicitation sessions — the author’s original discovery conversations with Copilot, conducted well before this write-up — produced geometrically structured, non-trivial descriptions the author judged internally consistent and reusable as design guidance. The formal acceptance criteria (independent-session reproducibility verification, primitive-vocabulary comparison) are protocol-defined and pending execution once session records are assembled into ART-001 (Section 8). The reproducibility criterion is treated as unverified pending that archival assembly; only receipted findings are reported.

What this establishes: the elicitation procedure produces a reproducible generative artifact with stable structure — a usable scaffold for formalization. **What this does not establish:** that the descriptions correspond to the model’s actual internal geometry. The reproducibility observed is a property of the model’s generative distribution under geometric framing prompts, consistent with its training exposure to representational-geometry literature (Section 2.4). Both sentences belong in any citation of this finding.

5.2 Structural Manifold Properties

The three well-formedness properties of Section 3.2 — continuity, in-principle differentiability, and parametric reproducibility — are defined as acceptance criteria for the ART-002 artifact. Because ART-002 has not been produced (Section 8), these criteria are protocol-defined and untested. The decoupling property this section was meant to evidence — downstream phases executing without the original model — remains a design objective of the certification roadmap, not an achieved result.

5.3 Blueprint Reproducibility

The reproducibility protocol for the DAB is defined as three independent execution runs: fresh environment import, full assembly from the blueprint, and Phase 5 consistency testing per instance.

Status: protocol defined; the three-run independent re-execution trial is scheduled as part of the upcoming 60-day benchmark battery and will be receipted in ART-005 when executed. Blueprint portability across a second independent builder instance is likewise protocol-defined and pending the same trial. Figures reported for these trials in earlier drafts were not backed by measurement records and are withdrawn pending execution.

5.4 Determinism Evidence — Substrate Level (VERIFIED)

The verified determinism evidence in this program is substrate-level and receipted. **Substrate Determinism Receipt 1 (SDR-1)**, certified September 2026, established the first certified deterministic substrate with byte-identical replay parity across three phases:

Phase	Test	Result
A — Parity	Identical outputs across parallel substrate instances	PASSED
B — Replay	Byte-identical output on replay of recorded inputs	PASSED
C — Temporal Independence	Identical outputs regardless of execution timing or instance index	PASSED

Verification checksum: **0xc5f88dcf**. The deterministic substrate produces identical outputs regardless of execution timing or instance index — this is the receipted, auditable foundation on which the dual-layer runtime operates, and it is the strongest determinism claim this document makes.

Operationally, the runtime instances built on this substrate have been exercised continuously by scheduled nightly chaos-training cycles (11:00 PM CT), which inject synthetic anomalies and record healing outcomes against a measured auto-heal baseline of 88.5%. These cycles constitute the ongoing evidence-gathering mechanism for runtime behavior under perturbation, and their receipts are maintained in the append-only ledger.

5.5 Runtime-Layer Metrics (Protocol Defined — Execution Pending)

The following Phase 5 metrics are defined and instrumented, with results to be recorded in ART-005 upon execution during the scheduled benchmark battery:

Metric	Definition	Target	Result
Determinism Score	Percentage of input runs producing identical outputs across N trials (N recorded)	100%	PENDING — scheduled
Constraint Adherence Rate	Constraint checks passed without violation in defined domain	100%	PENDING — scheduled
Layer 1–2 Sync Latency (max)	Maximum observed latency between Layer 1 event and Layer 2 transition	< 10 ms	PENDING — scheduled
Cross-Layer Orphan Events	State transitions with no corresponding Layer 1 classification event	0	PENDING — scheduled
Independent Execution Runs	Independent DAB executions yielding identical runtime behavior	3 / 3	PENDING — scheduled

Evidence note: figures reported for these metrics in earlier drafts were not backed by measurement records and are withdrawn pending the scheduled trials. This program reports verified receipts and pending protocols — nothing else.

6. Discussion

6.1 Significance

The principal contribution of this work is methodological: a structured, artifact-producing pipeline that converts a model-generated geometric narrative into a formal specification, and from that specification into a deployed, auditable, deterministic runtime. The pipeline does *not* demonstrate that the model's self-description reflects its internals (it does not, and cannot — Section 2.4). What it demonstrates is that a disciplined formalization chain — narrative → SMS → DAB → deployed runtime → receipted validation — can carry an engineering program from open-ended elicitation to certified deterministic behavior with full artifact provenance.

The structural manifold functions as a stable, decoupling intermediate representation: it separates the discovery phases (1–2, which depend on the model's generative behavior) from the engineering phases (3–5, which depend only on the SMS artifact). Once produced and validated, the downstream phases are executable independently of the original LLM — essential for certification, since the runtime can be audited and reproduced without continued access to the original model instance.

6.2 Implications for Certification (RDC-1)

The pipeline is structured to satisfy the chain-of-provenance requirements of the RDC-1 (Runtime Determinism Certification, Level 1) framework: each phase produces a named, versioned, format-specified artifact, and each artifact is an auditable checkpoint in the chain from initial prompt to deployed runtime.

Provenance Step	Artifact	Certifiable Property	Evidence Status
Elicitation	ART-001: Geometric Self-Description	Prompt reproducibility; representation consistency (of the artifact, not the model — §2.4)	Sessions complete; archival audit pending
Formalization	ART-002: Structural Manifold Specification	Internal consistency; parametric stability	Not produced — procedure defined
Blueprint	ART-003: Deterministic Architecture Blueprint	Completeness; portability; version control	Not produced — procedure defined
Execution	ART-004: Runtime Instance	Blueprint fidelity; no implementation drift	Deployed; conformance re-verification scheduled
Validation	ART-005: Validation Report	Determinism; constraint adherence; layer synchronization	Substrate determinism VERIFIED (SDR-1, 0xc5f88dcf); runtime metrics pending

6.3 Considerations and Future Work

- **Cross-model generalization:** the structural manifold is model- and session-specific. Extending the pipeline to additional model families is required to establish whether the formalism generalizes and whether a common manifold vocabulary can be defined across architectures.
- **Automated manifold formalization:** Phase 2 currently requires manual formalization. Automated tooling to parse elicitation documents and generate SMS artifacts would reduce time and improve consistency.
- **Runtime monitoring integration:** live monitoring surfaces for Layer 1 and Layer 2 behavioral metrics, fed by the existing audit logs, would support ongoing operational certification maintenance.
- **Manifold coverage expansion:** the SMS captures only the structures surfaced by the prompts used. A systematic prompt-coverage study is required to bound the completeness of the specification.
- **Execute the pending trials:** the three-run DAB re-execution trial and the full Phase 5 metric suite (Section 5.5) should be executed and receipted before this document is updated from draft to final; evidence first, status second.

7. Conclusion

This document has presented a complete technical account of a five-phase pipeline that begins with the structured elicitation of geometric self-descriptions from a large language model and concludes with a deployed dual-layer runtime operating on a verified deterministic substrate. Phase 1 — the author’s original discovery sessions — produced geometrically structured narrative (pending consolidation into ART-001), treated throughout as scaffold, not introspective measurement. Phases 2 and 3 (the SMS and DAB formalization artifacts) are defined methodology, not yet executed; their statuses match the artifact inventory (Section 8). Phase 4 instantiated the runtime on Base44 without manual code authoring (ART-004 — real, deployed, in prototype service). Phase 5 defines the validation protocol, with substrate-level determinism verified and receipted (SDR-1, checksum 0xc5f88dcf) and runtime-layer metrics scheduled for measurement (ART-005).

The key methodological contribution is the structural manifold as a stable, decoupling intermediate representation — one that separates the model-dependent discovery phase from the engineering and validation phases, enabling the latter to be executed and audited independently. The key epistemic discipline, added in this revision, is the refusal to let narrative stand in for evidence: the model’s story is where the pipeline begins, not what it proves. What it proves is the receipted, byte-identical behavior of the substrate and the auditable artifact chain that produced the runtime.

This package is submitted for internal review as RDC-1 certification reference material, pending execution and receipting of the scheduled validation trials described in Section 5.5.

8. Artifact Inventory

Artifact ID	Artifact Name	Phase	Format	Status (v1.1, audited)
ART-001	Geometric Self-Description Document	Phase 1	Markdown / PDF	Discovery sessions are real (author's original conversations with Copilot); consolidated document does not exist yet — assembly from session records pending
ART-002	Structural Manifold Specification (SMS)	Phase 2	JSON Schema	Not yet produced — procedure defined; execution pending on the certification roadmap
ART-003	Deterministic Architecture Blueprint (DAB)	Phase 3	YAML / JSON Schema	Not yet produced — procedure defined; execution pending on the certification roadmap
ART-004	Runtime Instance	Phase 4	Deployed Application	Deployed and operating in prototype service; DAB-conformance re-verification scheduled with Phase 5 trials
ART-005	Dual-Layer Runtime Validation Report	Phase 5	PDF (receipted)	Substrate determinism verified (SDR-1 receipt); runtime-layer metrics pending scheduled execution

Inventory note: statuses are maintained under a receipt-first discipline — an artifact is marked complete only when it exists and can be produced for audit. Verified reality: ART-004 (deployed Base44 runtime instances) and ART-005's substrate component (SDR-1 receipt, checksum 0xc5f88dcf). ART-002 and ART-003 are defined-but-unexecuted; ART-001 exists as discovery sessions pending consolidation into a single document.

9. Glossary

Deterministic Architecture Blueprint (DAB) — the versioned, machine-readable specification that fully describes the runtime architecture; the single source of truth for Phase 4 execution.

Manifold Hypothesis — the principle in representation learning holding that high-dimensional natural data tends to concentrate near lower-dimensional manifolds embedded in the ambient space; theoretical justification for treating LLM activation spaces as exhibiting manifold-like geometric structure.

Narrative Scaffold (v1.1) — the role assigned to the LLM's geometric self-description: a reproducible generative artifact used as the origin of a formal specification chain, explicitly not a claim of introspective access to model internals.

RDC-1 (Runtime Determinism Certification, Level 1) — the certification framework for which this document is a primary technical reference, specifying chain-of-provenance, auditability, and determinism requirements for runtimes deployed in environments requiring formal runtime assurance. *Renamed in v1.1 to eliminate the collision with SDR-1 below.*

Reproducibility Criterion — the formal standard applied to the DAB: a blueprint satisfies the criterion if, applied to a compatible builder environment, it produces a runtime whose input/output behavior is functionally identical to the original across all inputs in the defined domain.

SDR-1 (Substrate Determinism Receipt 1) — a completed, verified determinism test result (September 2026): the deterministic substrate produced byte-identical outputs (checksum 0xc5f88dcf) across parity, replay, and temporal-independence phases. SDR-1 is evidence, cited in Section 5.4 — not a certification framework, and not this document.

Semantic Interpretation Layer — Layer 1 of the dual-layer runtime: parsing, classifying, and routing incoming signals per the semantic schema derived from the structural manifold, recorded in classification and routing logs.

Structural Manifold — a formalized geometric abstraction of operational topology, produced through controlled elicitation and formalization; an engineering construct, not a claim about model internals.

Execution Control Layer — Layer 2 of the dual-layer runtime: executing state transitions, enforcing constraints, and producing outputs per the DAB, recorded in the execution audit trail.

10. Appendix A: Sample Prompt Templates

The following templates are representative of the three prompt categories employed in Phase 1. They are generic and may be adapted to other model targets. Each is designed to elicit geometric, spatial, or topological content through open-ended framing.

A.1 Topology Elicitation Prompt

Purpose: to surface a description of the overall spatial organization of the processing — the shape of the operational space, its regions and boundaries, and the connectivity of its principal zones.

If you were to describe the overall shape or structure of the space through which information moves as you process a complex input, what would that space look like? Consider not just a single pass from input to output, but the full topology of the region — its boundaries, its connected zones, any areas where information tends to concentrate or disperse, and any structural features that feel persistent across different types of inputs. Describe this space in as much geometric or spatial detail as you find natural, using whatever metaphors or formal vocabulary feels most accurate to your experience of the process.

A.2 Dimensionality Self-Assessment Prompt

Purpose: to elicit a characterization of the effective dimensionality of the representational spaces — the number and nature of independent axes of variation across processing regimes.

When you consider the internal representations you work with as you process information — the structures that carry meaning from one stage of processing to the next — how many independent dimensions or axes of variation do those representations seem to have? Does that number feel consistent across all types of inputs, or does the effective dimensionality seem to change depending on the nature of the input or the stage of processing? Describe, as precisely as you can, the dimensional character of the representational space you occupy, including any regions where the space feels higher or lower dimensional than its surroundings.

A.3 Relational Geometry Prompt

Purpose: to surface a description of the relational and directional structure between distinct processing elements — proximity, distance, connection, and flow.

Consider the different functional regions or components involved in your processing of an input. If you were to place these regions in a shared geometric space and describe their spatial relationships to one another — which are close, which are distant, which are connected by direct pathways and which are separated by boundaries — what would that relational map look like? Describe how information moves between these regions: is the flow directional, bidirectional, or recursive? Are there regions that act as hubs through which information must pass, or corridors that channel information along specific trajectories? Describe the relational geometry of these components in as much spatial and directional detail as you are able to produce.

Usage note: these templates are starting points. Follow-up probes should be applied to responses satisfying the Section 4.1 validity criteria to extract additional geometric precision. All prompts and responses used in the actual Phase 1 sessions are preserved in ART-001. Per Section 2.4, outputs of these prompts are narrative artifacts; their acceptance criteria are consistency, reproducibility, and formalizability — not correspondence to model internals.

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